1. Import the library

```
from sklearn.model_selection import train_test_split
```

2. Separate features and target

Suppose your DataFrame is named `df` and the target column is `target`.

```
X = df.drop("target", axis=1)   # Features
y = df["target"]                # Target
```

3. Split into training and testing data

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,      # 20% for testing
    random_state=42,    # Ensures reproducibility
    shuffle=True
)
```

4. Check the shapes

```
print("X_train shape:", X_train.shape)
print("X_test shape :", X_test.shape)
print("y_train shape:", y_train.shape)
print("y_test shape :", y_test.shape)
```

Complete Example

```
import pandas as pd
from sklearn.model_selection import train_test_split

# Load dataset
df = pd.read_csv("dataset.csv")

# Features and Target
X = df.drop("target", axis=1)
y = df["target"]

# Train-Test Split
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

print("Training Data")
print(X_train.head())
```

```
print("\nTesting Data")
print(X_test.head())
```

If your target column is named differently

For example, if your target column is `Price`:

```
X = df.drop("Price", axis=1)
y = df["Price"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.25,
    random_state=42
)
```

```
# Import libraries
import pandas as pd
from sklearn.model_selection import train_test_split

# Load dataset
df = pd.read_csv("blinkit_data.csv")

# View first 5 rows
print(df.head())

# Check missing values
print(df.isnull().sum())

# --------------------------------
# Features (X) and Target (y)
# --------------------------------

X = df.drop("Rating", axis=1)   # Remove target column
y = df["Rating"]                # Target column

# --------------------------------
# Train-Test Split
# --------------------------------

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,      # 20% Test Data
    random_state=42,
    shuffle=True
)

# --------------------------------
# Display Shapes
# --------------------------------

print("X_train Shape :", X_train.shape)
```

```
print("X_test Shape  :", X_test.shape)
print("y_train Shape :", y_train.shape)
print("y_test Shape  :", y_test.shape)
```

If your dataset contains categorical columns

Encode them before training:

```
from sklearn.preprocessing import LabelEncoder

le = LabelEncoder()

for col in X.select_dtypes(include="object").columns:
    X[col] = le.fit_transform(X[col].astype(str))
```

Then split the data:

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
```

If you also want feature scaling

```
from sklearn.preprocessing import MinMaxScaler

scaler = MinMaxScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

Complete ML Pipeline

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, MinMaxScaler

# Load dataset
df = pd.read_csv("blinkit_data.csv")

# Features and Target
X = df.drop("Rating", axis=1)
y = df["Rating"]

# Encode categorical columns
le = LabelEncoder()
for col in X.select_dtypes(include="object").columns:
    X[col] = le.fit_transform(X[col].astype(str))

# Train-Test Split
X_train, X_test, y_train, y_test = train_test_split(
```

```
    X,
    y,
    test_size=0.2,
    random_state=42
)

# Scale features
scaler = MinMaxScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

print("Training Data Shape:", X_train.shape)
print("Testing Data Shape :", X_test.shape)
```